

NLP-based Gujarat Smart Reply System (Extension) for WhatsApp Web

Subject - Natural Language Processing (NLP)

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Fair & Equal Contribution Breakdown

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Data Collection & Preprocessing
of Gujarati chat data

Feature Engineering (Sparse
character-level TF-IDF)

Fine-tuning Custom FastText Embeddings

Training and optimization of
the LinearSVC model

Param Maru

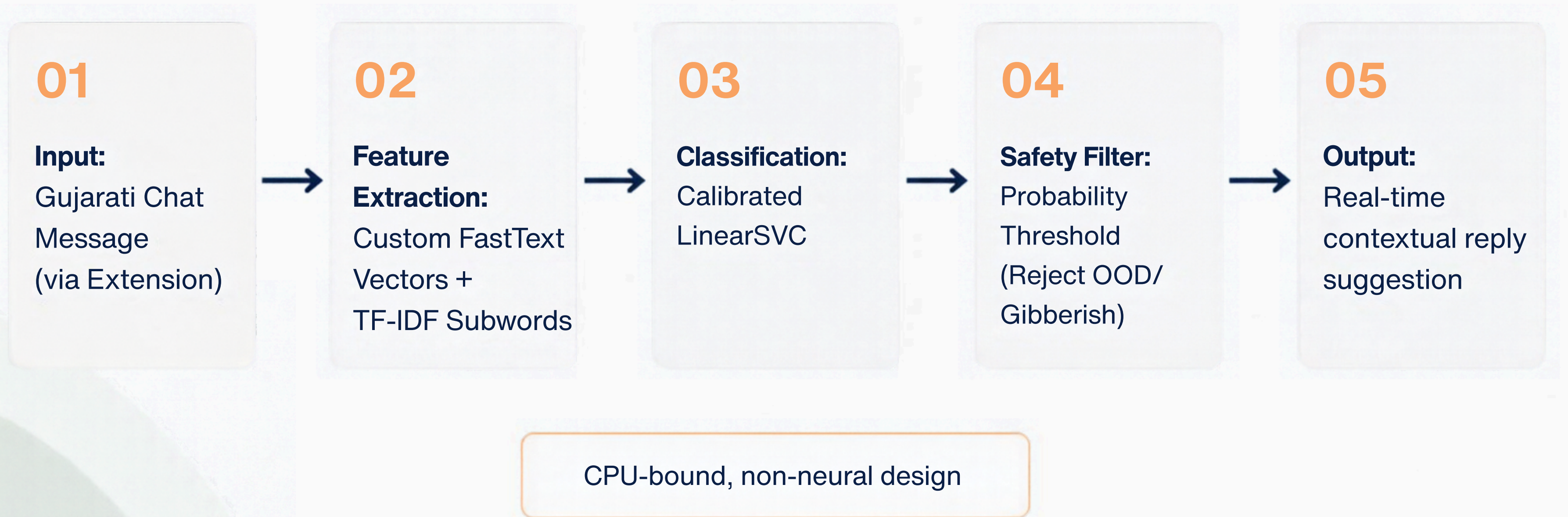
Local Flask API Backend Development

Out-of-Domain (OOD) rejection
mechanism (probability thresholding)

Chrome Extension Integration for
real-time inference

System latency measurement &
operational metrics analysis

Proposed Methodology



Experiment & Results

Operational Performance

Key operational metrics of the Gujarati smart reply system.

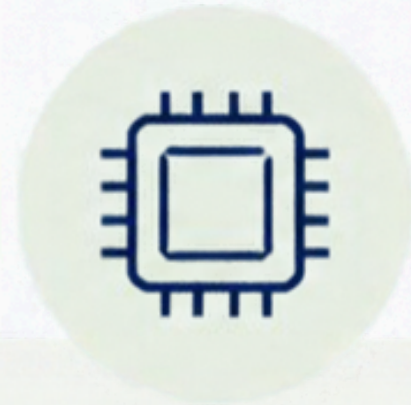


Inference Latency

0.77 ms

per request

sub-millisecond speed



Memory Footprint

805.43 MB

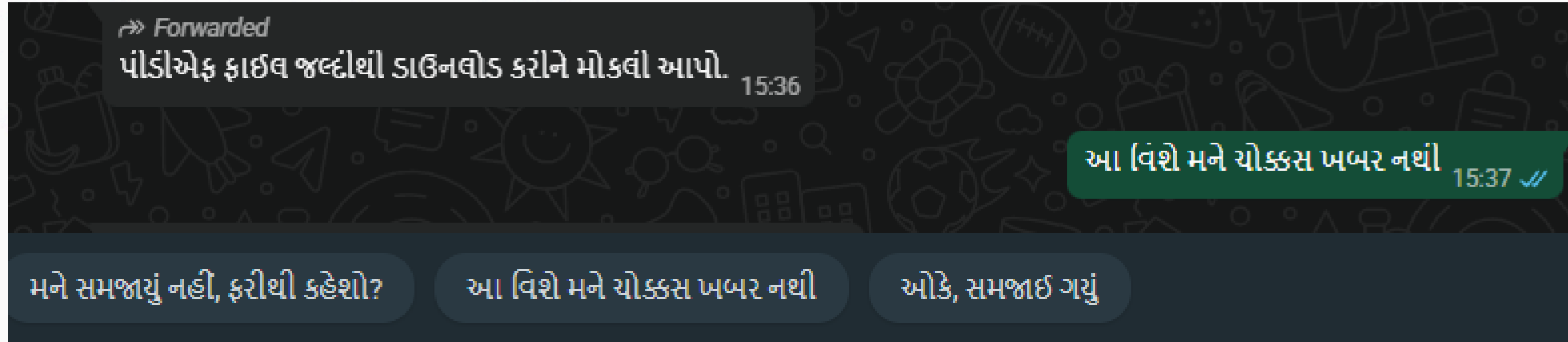


Corpus Ingestion Time

~120

Seconds

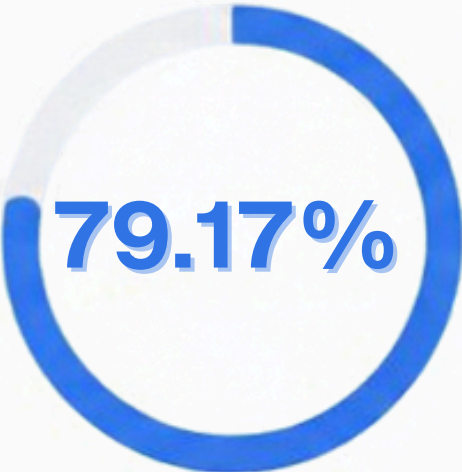
Working of Extension



The Chrome extension captures incoming WhatsApp Web messages and sends them to a local Flask backend, where an NLP model (FastText + TF-IDF + LinearSVC) classifies intent. It then injects clickable, contextual Gujarati reply buttons directly above the chat box for instant, one-tap messaging.

Intent Classification & System Robustness

Intent Classification



Overall Accuracy



Macro F1-Score



Weighted F1-Score



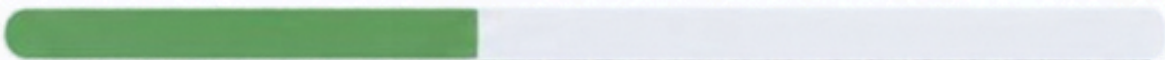
Macro Precision

Highlights low false positives

Calibration & Safety

Mean Confidence Score (In-Domain)

0.3534



OOD Rejection Rate

50.00%



Highlights success at threshold 0.25

Literature Survey

● **Kannan et al. (2016)**

Foundation of automated response suggestion for email

● **Henderson et al. (2017)**

Efficient natural language response suggestion

● **Ji et al. (2014)**

Information retrieval approach to short text conversation

● **Wu et al. (2017)**

Sequential matching network for multi-turn response selection

● **Kakwani et al. (2020)**

IndicNLPSuite for Indian language pre-trained models

● **Shang et al. (2015)**

Neural responding machine for short-text conversation



References

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- [2] Henderson, M., et al. (2017). Efficient natural language response suggestion for smart reply. *arXiv*.
- [3] Ji, Z., Lu, Z., & Li, H. (2014). An information retrieval approach to short text conversation. *arXiv*.
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- [5] Kakwani, D., et al. (2020). Indicnlp suite: Monolingual corpora, evaluation benchmarks and tools for indian languages. *EMNLP*.
- [6] Shang, L., Lu, Z., & Li, H. (2015). Neural responding machine for short-text conversation. *ACL*.
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